

Once we have an image we can run commands within a container managed by Docker.

```
docker::run [ 'collabnix-puppet':
image => 'ajeetraina/collabnix',
commands => '/usr/sbin/apache2 -D FOREGROUND',
ports => '80',
]
```

The first uncommented statement is *include docker module*. The statement `docker::image { 'ubuntu': image_tag: 'precise'}` is equivalent to running the `$ docker pull ubuntu:precise` command in the console. Considering that the *ajeetraina/collabnix* image carries Apache2 packages, it will try to run the *httpd* service.

The above code is similar to Dockerfile:

```
#cat Dockerfile
FROM Ubuntu:precise
MAINTAINER Ajeet Raina ajeetraina@gmail.com
RUN yum install -y httpd
RUN chkconfig httpd on
```

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```
struct sockaddr_l2 addr;
char dest[18] = "<Bluetooth Address>" //put server's
bluetooth adapter address you can find address from command
hciconfig -a.
addr . l2_family = AF_BLUETOOTH ;
addr . l2_psm = htobs(0x1001 ) ;
str2ba ( dest , &addr . l2_baddr ) ;
connect ( s , ( struct sockaddr * )&addr , sizeof ( addr ) ) ;
```

Now you can send data to the server:

```
if ( check == 0 ) {
status = send ( s , "Hi Subodh " , 10 , 0 ) ;
}
```

Pairing with a Bluetooth-enabled mobile

Your mobile also has a server so that it can connect to a client, and send and receive files. For pairing your system with a mobile, first, you have to open a socket:

```
s = socket ( AF_BLUETOOTH, SOCK_STREAM, BTPROTO_RFCOMM);
```

Now connect to the Bluetooth address of the mobile:

```
struct sockaddr_rc addr;
char dest[18] = "<Mobile Bluetooth Address>" ;
addr . rc_family = AF_BLUETOOTH ;
addr . rc_psm = 1 ;
```

EXPOSE 80

```
CMD ["/usr/sbin/apache2", "-D", "FOREGROUND"]
```

The above Puppet manifest entry should be added to *etc/puppet/manifests/site.pp* for the Puppet agent node. Remember that `docker::run` takes care of the build process and, therefore, the container can be run flawlessly.

Hence, the above Puppet module automates the creation of a Docker container, and installs and configures the services on the Puppet agent nodes successfully. 

References

- [1] <https://forge.puppetlabs.com/garethr/docker>
- [2] <https://docs.docker.com/articles/puppet/>
- [3] <https://puppetlabs.com/presentations/using-docker-puppet-james-tumbull-kickstarter>

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```
str2ba ( dest , &addr . rc_baddr ) ;
connect ( s , ( struct sockaddr * )&addr , sizeof ( addr ) ) ;
```

As you run this program, your mobile server will receive a bonding accepting message.

Accept this connection and send the encryption key.

After sending the key from the mobile, your system will show a pairing confirmation window. Enter the same number that you entered in your mobile

You have now connected with the mobile's server.

As an exercise, try to implement a Bluetooth headset profile (HSP). 

References

- [1] <http://people.csail.mit.edu/albert/bluez-intro/c404.html#lbtz-choosing>
- [2] http://en.wikipedia.org/wiki/Bluetooth_stack
- [3] <http://www.drdoobs.com/mobile/using-bluetooth/232500828?pgno=1>
- [4] http://en.wikipedia.org/wiki/List_of_Bluetooth_protocols
- [5] http://www.eetimes.com/document.asp?doc_id=1275960
- [6] <http://en.wikipedia.org/wiki/Bluetooth>
- [7] <https://www.wusewji-linux-books.googlecode.com/files/Essential%20Linux%20Device%20Drivers%5B2008,851p%5D.pdf>

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